

## 第五章 作业题2

已知两个二进制数的真值分别为  $X=-1101$ ， $Y=-11$ ，用补码一位乘法公式求 $X \times Y$ ，给出完整的过程

$$[X_{\text{补}}] = 10011 \quad [-X_{\text{补}}] = 01101 \quad [1]_{\text{补}} = 11101$$

$$\begin{array}{r} 000000 \\ 001101 \\ \hline \end{array}$$

$$\begin{array}{r} 001101 \\ 000110 \\ 110011 \\ \hline \end{array}$$

$$\begin{array}{r} 111001 \\ \hline \end{array}$$

$$\begin{array}{r} 111100 \\ \hline \end{array}$$

$$\begin{array}{r} 001101 \\ \hline \end{array}$$

$$\begin{array}{r} 001001 \\ \hline \end{array}$$

$$\begin{array}{r} 000100 \\ \hline \end{array}$$

$$0$$

$$\begin{array}{r} 000100 \\ \hline \end{array}$$

$$111010$$

$$1111010$$

$$1111010$$

$$1111010$$

$$\begin{array}{cccccccc|cccc}
 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\
 0 & & & & & & & & & & & & & 
 \end{array}$$


---


$$\begin{array}{cccccccc|cccc}
 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 1 & & & 
 \end{array}$$

$$X'f = 1 \ 0 \ 0 \ 1 \ 1 \ 1$$